

Performance in Bull/Bear Scenarios, and Pros and Cons of the Strategy

Performance Across Market Conditions

In sports betting, "bull" and "bear" don't map perfectly to financial markets, but there are clear analogs. A "bull" condition for this strategy is when there are many matches with clear statistical mismatches that the public is mispricing. A "bear" condition is when the market is efficient, matches are closely contested, and there's little edge to extract.

Bull conditions (where the strategy thrives):

The strategy performs best during the group stage of major tournaments, where casual betting volume is highest and narrative bias is strongest. The 2022 World Cup group stage is a good example. You have 48 matches across 16 groups, many featuring lopsided matchups (e.g., Spain vs. Costa Rica, England vs. Iran) where the model can confidently predict outcomes from player statistics. But more importantly, you also get matches where the public massively overvalues a famous team and the underlying player data tells a different story. Our model's 52.3% overall accuracy on the 2022 World Cup was achieved across both types of matches, and the memorization test (54.7% on anonymized vs. 50.0% on named teams) suggests the model is actually better at extracting signal from raw stats than from name recognition. This is exactly the profile you want: the model's edge comes from situations where the "brand" of a team diverges from the current reality of its squad composition.

The strategy also benefits from periods of high squad turnover. Between major tournaments, national teams can look very different due to retirements, injuries, and emerging talent. Historical team-level models (Elo, pi-ratings) are slow to adapt to these shifts because they're based on match results that may reflect a completely different roster. Our player-level approach updates automatically as soon as the new squad data is available, which gives it a structural timing advantage.

Bear conditions (where the strategy struggles):

Knockout stage matches between closely-ranked teams are the toughest environment. When Argentina plays France in a World Cup Final, the player statistics on both sides are excellent, and the margin of victory often comes down to individual moments (a penalty, a red card, a goalkeeper save) that no statistical model can predict. In these matches, the model's predicted probabilities tend to be close to 50-50, which means there's very little edge even if the prediction is correct. The strategy's trading rules help here by requiring a minimum edge threshold, which naturally filters out these low-edge matches. But it means the strategy sits idle during the most high-profile (and highest-liquidity) matches of the tournament.

Draw prediction is another bear scenario. Our model, like every football prediction model in the literature, struggles with draws. Draws represent about 25% of World Cup outcomes but are driven by situational factors (defensive tactics, fatigue, weather, late equalizers) that are poorly captured by pre-match player statistics. The best XGBoost models in the academic literature achieve only about 30% F1 score on draws, and our LLM is no different. This is a structural limitation of the data, not the model.

Pros

Interpretability. This is the single biggest advantage over any other approach in the literature. Traditional ML models for football prediction (XGBoost, CatBoost, random forests, logistic regression) output a probability with no explanation. Our model outputs a reasoning chain that references specific

player statistics. You can read the reasoning before placing a bet, catch factual errors, and understand why the model is making a particular prediction. In the Prophet Arena benchmark for LLM-based forecasting, the researchers found that LLM reasoning quality varies dramatically by domain and that the ability to inspect reasoning is what separates useful predictions from noise. Our approach makes this inspection the default.

Player-level granularity. Most football prediction models in the literature operate at the team level, using features like Elo ratings, pi-ratings, or aggregate team statistics. The best-performing model in the 2017 Soccer Prediction Challenge (CatBoost + pi-ratings, 55.82% accuracy, 0.1925 RPS) used team-level rating features, not player data. Our approach goes one level deeper by reasoning from the actual starting XI and their individual statistics across domestic leagues. This means the model can in principle capture information that team-level models miss, like the impact of a key player returning from injury or a squad being weakened by suspensions. Whether it actually does this well enough with only 384 training samples is an open question, but the architectural capacity is there.

Memorization resistance. The anonymized variant test showed the model performs at least as well on "Team A vs. Team B" samples as on named team samples. This is a form of robustness that most prediction systems never bother to validate. It means the model's predictions are grounded in the input statistics, not in prior knowledge about which teams are historically strong. This matters because team reputations are sticky but squad quality is not.

Transferability. The prompt-based architecture means the same model can be applied to any football competition (leagues, cups, international tournaments) without retraining, as long as the player data is available in the same format. Traditional ML models trained on World Cup data typically cannot be directly applied to league matches without feature re-engineering and retraining.

Cons

Small training set. 384 samples is tiny by ML standards. The 2017 Soccer Prediction Challenge used 216,000+ matches for training. Our model learned from three World Cups (192 matches, doubled to 384 with anonymized variants). While fine-tuning a pretrained LLM requires less data than training from scratch, the model has seen a very narrow slice of football. It has never seen a Premier League match, a Champions League match, or any domestic league game during training. Whether it generalizes to those formats is unvalidated.

No dynamic Elo or rating features. The academic literature consistently shows that Elo-based rating features are the strongest predictors for football match outcomes. Our model does not use Elo ratings. It uses raw player statistics, which are informative but miss the "momentum" and "form" signals that Elo captures (recent results, strength of schedule, home/away performance trajectory). Adding Elo or pi-ratings as additional context in the prompt would likely improve performance.

No calibration optimization. A recent study on ML for NBA betting (Machine Learning with Applications, 2025) demonstrated that model calibration is more important than accuracy for profitable sports betting. A well-calibrated model that says "60% probability" and is right 60% of the time enables Kelly sizing to work correctly. An overconfident model that says "75%" but is right 55% of the time will destroy a Kelly-based bankroll. We measured accuracy but not calibration. Our 128-sample eval set is likely too small for a meaningful calibration analysis, but this is a critical gap.

Compute cost and inference speed. Running an 8B parameter model for each prediction is far more expensive than running XGBoost (milliseconds vs. seconds). This doesn't matter for pre-match prediction

on a few thousand matches per year, but it makes the approach impractical for any scenario requiring rapid iteration (live in-play prediction, parameter sweeps across thousands of simulations).

No real-time data integration. The model predicts from a static snapshot of player statistics. It does not incorporate late-breaking information like lineup changes, injuries announced on match day, weather conditions, or real-time odds movements. Professional sports bettors and bookmakers adjust their models continuously as new information arrives. Our strategy relies on the human reasoning audit to catch some of this (reviewing the reasoning for injured players), but that's a manual process and won't catch everything.